

Recursive Sandhi and Vichcheda Canonicalization for Exterior-Product Contractions

Akash Dwivedi

akash.d.dwivedi@gmail.com

<https://www.linkedin.com/in/akash-dvd-3259402a/>

October 7, 2026

Abstract

Symbolic exterior-algebra calculations must preserve both antisymmetry and the scalar factors created when left-contraction panels share geometric factors. This paper studies recursive stitching for contraction panels whose down arguments are decomposable blades. The local grade- r seam identity is proved by an iterated-contraction minor expansion: a grade- r contractor can be stitched across a grade- r seam with scalar coefficient given by their Capelli pairing. Nested stitching accumulates contractor and seam blades together, while a grade-balance gate determines whether to descend, emit a scalar, or vanish under the sole-seam invariant.

The local identity and the fixed-schedule induction theorem are proved for the well-formed panel grammar using a level-local visibility closure lemma. Schedule-independent normalization remains open; cases through nesting depth six have been checked by direct symbolic expansion.

The grade- r identity, its vanishing branch, the Capelli coefficient, a closed-form nested identity of arbitrary depth, and the up-vichcheda vanishing rule are machine-checked in Lean 4 with Mathlib, over any commutative ring and any bilinear form.

Keywords: geometric algebra; exterior algebra; left contraction; symbolic canonicalization; *sandhi*; *vichcheda*; Capelli; normal form

Introduction

Exterior products provide a compact representation of antisymmetric multi-vectors, but symbolic expressions become difficult to manipulate when left-contraction panels are wedged together and share factors. The elementary pattern is

$$(a \lrcorner (b \wedge s)) \wedge (a \lrcorner (s \wedge c)).$$

The shared factor s is the seam. The base identity combines the panels as

$$(a \cdot s) a \lrcorner (b \wedge s \wedge c).$$

The sign is determined by orienting the seam toward the join.

This paper develops the grade-matched generalization in which a grade- r contractor meets a grade- r seam blade. The local identity is proved by an iterated-contraction minor expansion, and nested *sandhi* is formulated as repeated application of this local theorem. The contractor and seam grow together so that their pairing remains scalar.

The local grade- r identity and the fixed-schedule induction theorem are proved for the well-formed panel grammar using a level-local visibility closure lemma. Schedule-independent normalization remains open; cases through nesting depth six have been checked by direct symbolic expansion.

The stitching identities themselves are machine-checked. The grade- r identity, its vanishing branch, the Capelli coefficient, and a closed-form nested identity of arbitrary depth (Theorem 3) are proved in Lean 4 with Mathlib, over any commutative ring and any bilinear form (Section 9); the Lean source is at <https://github.com/Akash-dvd/sygal/tree/main/paper/sandhi/lean>. Locating a seam inside a given expression is a separate search problem, solved by the implementation and not part of these theorems.

1 Setting

Let V be a vector space with a nondegenerate bilinear form. We use $\wedge$ for the exterior product, $\lrcorner$ for left contraction, and $|$ for the scalar pairing of equal-grade blades.

For a vector u ,

$$u \lrcorner (X \wedge Y) = (u \lrcorner X) \wedge Y + (-1)^{\text{grade}(X)} X \wedge (u \lrcorner Y).$$

For a decomposable blade

$$A = a_1 \wedge \cdots \wedge a_r,$$

left contraction is the ordered iteration

$$A \lrcorner X = a_1 \lrcorner (a_2 \lrcorner (\cdots (a_r \lrcorner X) \cdots)).$$

Equivalently,

$$(u \wedge A) \lrcorner X = u \lrcorner (A \lrcorner X).$$

This recursive rule fixes the contraction-order sign convention. The input fragment consists of exterior products of contraction panels. Their ordering is part of the exterior-algebra data: reordering factors contributes parity signs, and repeated factors can make an expression zero.

2 Base Identity

Let a be a vector and let A, B be decomposable blades with one distinguished shared one-vector s . After reordering their factors, write

$$A = (-1)^{\sigma_A} U \wedge s, \quad B = (-1)^{\sigma_B} s \wedge V, \quad \sigma_A, \sigma_B \in \{0, 1\},$$

where U and V have no common factors. Define the oriented concatenation

$$A \sqcup B := (-1)^{\sigma_A + \sigma_B} U \wedge s \wedge V.$$

Proposition 1 (Sole-shared-seam identity). *Under these hypotheses,*

$$(a \lrcorner A) \wedge (a \lrcorner B) = (a \cdot s) (a \lrcorner (A \sqcup B)).$$

Proof. Put $p = \text{grade}(U)$ and $\epsilon = (-1)^{\sigma_A + \sigma_B}$. The graded Leibniz rule gives

$$\begin{aligned} a \lrcorner A &= (-1)^{\sigma_A} ((a \lrcorner U) \wedge s + (-1)^p (a \cdot s) U), \\ a \lrcorner B &= (-1)^{\sigma_B} ((a \cdot s) V - s \wedge (a \lrcorner V)). \end{aligned}$$

Wedging and using $s \wedge s = 0$ yields

$$\begin{aligned} (a \lrcorner A) \wedge (a \lrcorner B) &= \epsilon (a \cdot s) [(a \lrcorner U) \wedge s \wedge V + (-1)^p (a \cdot s) U \wedge V \\ &\quad - (-1)^p U \wedge s \wedge (a \lrcorner V)]. \end{aligned}$$

The bracket is exactly the following Leibniz expansion:

$$\begin{aligned} a \lrcorner (U \wedge s \wedge V) &= (a \lrcorner U) \wedge s \wedge V \\ &\quad + (-1)^p U \wedge ((a \cdot s) V - s \wedge (a \lrcorner V)). \end{aligned}$$

Since $A \sqcup B = \epsilon (U \wedge s \wedge V)$, the claimed identity follows. $\square$

Remark 1. *The disjointness of U and V identifies s as the sole seam and ensures a nondegenerate concatenation. If U and V share a factor, then $U \wedge s \wedge V = 0$; the same expansion above shows that the left side also vanishes. Thus disjointness is needed for the sole-seam interpretation, not for the formal equality after repeated factors are allowed.*

3 Grade- r Theorem

The scalar in the recursive construction is determined by grade. Let A and S be decomposable blades of the same grade r , and write

$$B = \epsilon_B b \wedge S, \quad C = \epsilon_C S \wedge c, \quad \epsilon_B, \epsilon_C \in \{+1, -1\},$$

where the factors of b and c are disjoint from one another and from the displayed seam factors. Define

$$B \sqcup C := \epsilon_B \epsilon_C b \wedge S \wedge c.$$

Lemma 1 (Canonical orientation). *Fix any total order on atom labels, and extend it recursively (for example, lexicographically) to the immediate factors at the current panel node. For a panel with sole seam S , sorting its factors with the seam factors adjacent to the join gives a unique aligned form:*

$$B = \epsilon_B b \wedge S, \quad C = \epsilon_C S \wedge c,$$

where the displayed factors are in the fixed induced order. The coefficients $\epsilon_B, \epsilon_C \in \{+1, -1\}$ are the parities of the unique permutations from the original factor lists to these aligned lists.

Proof. The total order fixes the order of the residual factors and of the seam factors. The sole-seam hypothesis fixes which factors occupy the seam block and whether that block is placed at the right or left join. Hence each panel has one target list and one permutation to that list. Exterior antisymmetry contributes exactly the sign of that permutation. If repeated factors occur, the panel is zero and is outside the nonzero sole-seam case. $\square$

Theorem 1 (Grade- r sole-seam identity). *Under these hypotheses,*

$$(A \sqcup B) \wedge (A \sqcup C) = (A \mid S) (A \sqcup (B \sqcup C)).$$

Operational intuition (not part of the proof). Sort each panel using any fixed total order, turn the two seam blocks toward the join, and pay the two permutation parities. Stitch the aligned panels, then apply the grade gate: descend when the accumulated seam is still larger, emit the overlap scalar when grades balance, and return zero when alternation forces a repeated residual seam.

Proof. Write $A = a_1 \wedge \cdots \wedge a_r$ and $S = s_1 \wedge \cdots \wedge s_r$. For an ordered blade

$$X = x_1 \wedge \cdots \wedge x_m$$

and a position set $I = \{i_1 < \cdots < i_r\} \subseteq \{1, \dots, m\}$, put

$$X_I = x_{i_1} \wedge \cdots \wedge x_{i_r}, \quad X_{\widehat{I}} = x_1 \wedge \cdots \wedge \widehat{x_{i_1}} \wedge \cdots \wedge \widehat{x_{i_r}} \wedge \cdots \wedge x_m.$$

With the contraction convention of Section 1, define

$$\text{sgn}(I) := (-1)^{i_1 + \cdots + i_r - r}.$$

This is the sign of moving the selected factors to the contraction slots, combined with the reversal forced by the iterated left contractions. Thus

$$\Delta_A(I; X) := \text{sgn}(I) \det[(a_\alpha \cdot x_{i_\beta})]_{\alpha, \beta=1}^r$$

and the r -fold contraction is the explicit expansion

$$A \lrcorner X = \sum_{|I|=r} \Delta_A(I; X) X_{\widehat{I}}. \quad (3.1)$$

For $r = 1$, this gives the usual sign $(-1)^{i_1-1}$. For $I = \{1, \dots, r\}$, it gives the determinant pairing with the displayed ordered block, including the sign dictated by the contraction convention.

We now compare the coefficients of each residual wedge. Write

$$B = b_1 \wedge \dots \wedge b_p \wedge s_1 \wedge \dots \wedge s_r, \quad C = s_1 \wedge \dots \wedge s_r \wedge c_1 \wedge \dots \wedge c_q,$$

temporarily taking $\epsilon_B = \epsilon_C = 1$. Let $P \subseteq \{1, \dots, p\}$, $Q \subseteq \{1, \dots, q\}$, and $R \subseteq \{1, \dots, r\}$ describe a residual wedge

$$Y_{P,R,Q} := b_P \wedge s_R \wedge c_Q,$$

where each subscript retains the original order. A term on the left can produce this residual only by partitioning the residual seam indices as $R = T \dot{\cup} (R \setminus T)$: the factors s_T come from the first panel and $s_{R \setminus T}$ from the second. Define

$$\begin{aligned} I_{P,T} &:= (\{1, \dots, p\} \setminus P) \cup (p + (\{1, \dots, r\} \setminus T)), \\ J_{T,Q} &:= (\{1, \dots, r\} \setminus (R \setminus T)) \cup (r + (\{1, \dots, q\} \setminus Q)), \\ K_{P,R,Q} &:= (\{1, \dots, p\} \setminus P) \cup (p + (\{1, \dots, r\} \setminus R)) \\ &\quad \cup (p + r + (\{1, \dots, q\} \setminus Q)). \end{aligned}$$

Only choices with all three displayed index sets of cardinality r contribute. The seam shuffle has sign

$$\eta(R, T) := (-1)^{\#\{(u,v) \in T \times (R \setminus T) : u > v\}},$$

because $s_T \wedge s_{R \setminus T} = \eta(R, T) s_R$. Therefore the coefficient of $Y_{P,R,Q}$ on the left is

$$\sum_T \eta(R, T) \Delta_A(I_{P,T}; B) \Delta_A(J_{T,Q}; C). \quad (3.2)$$

The Laplace–Cauchy–Binet minor identity, applied to the two copies of the ordered seam block S , gives

$$\sum_T \eta(R, T) \Delta_A(I_{P,T}; B) \Delta_A(J_{T,Q}; C) = (A \mid S) \Delta_A(K_{P,R,Q}; b \wedge S \wedge c). \quad (3.3)$$

The sign comparison used in this convolution is the following elementary index calculation.

Lemma 2 (Sign in the minor convolution). *Let $[n] = \{1, \dots, n\}$, write $t = |T|$, $k = |R|$, and put*

$$h(R, T) := \#\{(u, v) \in T \times (R \setminus T) : u > v\}.$$

Let

$$\text{sgn}_S := (-1)^{\binom{r}{2}}$$

be the sign of the ordered seam block considered by itself. Then

$$\text{sgn}(I_{P,T}) \text{sgn}(J_{T,Q}) \eta(R, T) = (-1)^{p(k-t-q+|Q|)+h(R,T)} \text{sgn}_S \text{sgn}(K_{P,R,Q}). \quad (3.4)$$

Proof. Write

$$\begin{aligned} \sum I_{P,T} &= \sum ([p] \setminus P) + p(r-t) + \sum ([r] \setminus T), \\ \sum J_{T,Q} &= \sum ([r] \setminus R) + \sum T + r(q-|Q|) + \sum ([q] \setminus Q), \\ \sum K_{P,R,Q} &= \sum ([p] \setminus P) + p(r-k) + \sum ([r] \setminus R) \\ &\quad + (p+r)(q-|Q|) + \sum ([q] \setminus Q). \end{aligned}$$

Using $\text{sgn}(I) = (-1)^{\sum I-r}$, subtraction gives

$$\begin{aligned} & (\sum I_{P,T} - r) + (\sum J_{T,Q} - r) - (\sum K_{P,R,Q} - r) + h(R, T) \\ & \equiv p(k - t - q + |Q|) + \binom{r}{2} + h(R, T) \pmod{2}. \end{aligned}$$

Since $\text{sgn}_S = (-1)^{\binom{r}{2}}$, this is exactly (3.4). The first term in the remaining exponent counts the index shifts across the b -block and the c -block; $h(R, T)$ counts the seam shuffle needed to restore $s_T \wedge s_{R \setminus T}$ to s_R . These are precisely the cofactor and shuffle signs in the Laplace expansion. Hence each T -term has the sign on the right-hand side of (3.3), and the ordinary Laplace–Cauchy–Binet minor identity proves (3.3). $\square$

Thus (3.3) is not relying on an unspecified “prescribed” sign: the selected-index signs, block shifts, and seam shuffle have all been reduced to the explicit parity in (3.4).

3.1 The Laplace identity being used

The determinant result invoked here is the generalized Laplace expansion for complementary minors. For an $n \times n$ matrix M , and fixed row and column sets $I, J \subseteq \{1, \dots, n\}$ of size k ,

$$\det M = \sum_{|L|=k} (-1)^{\sum I + \sum L} \det M[I, L] \det M[I^c, L^c], \quad (3.5)$$

where the complementary row set is I^c , the complementary column set is L^c , and all minors use increasing index order. This is the generalized Laplace theorem, equivalently the Cauchy–Binet identity applied to compound matrices; see Gantmacher [1959, Ch. I, §5].

For the present rectangular pairing matrix, the exact compound-minor specialization is obtained by putting

$$[I]_Y := \det[(a_\alpha \cdot y_{i_\beta})]_{\alpha, \beta=1}^r.$$

With the b, S, c column blocks and the index sets defined above, (3.5) becomes

$$\sum_T \eta(R, T) \text{sgn}(I_{P,T}) \text{sgn}(J_{T,Q}) [I_{P,T}]_B [J_{T,Q}]_C = \text{sgn}_S \text{sgn}(K_{P,R,Q}) [K_{P,R,Q}]_{b \wedge S \wedge c} [S]_S. \quad (3.6)$$

The sign lemma verifies the conversion from the increasing-column minors in (3.5) to the contraction minors Δ_A . Since $A \mid S = \text{sgn}_S [S]_S$, equation (3.6) is exactly (3.3), not an additional heuristic sign assertion.

3.2 Recovery of the base case

Set $r = 1$, $p = q = 1$, $A = a$, $S = s$, $b = b$, and $c = c$. Then

$$a \lrcorner (b \wedge s) = (a \cdot b)s - (a \cdot s)b, \quad a \lrcorner (s \wedge c) = (a \cdot s)c - (a \cdot c)s.$$

Their wedge is

$$\begin{aligned} & (a \lrcorner (b \wedge s)) \wedge (a \lrcorner (s \wedge c)) \\ & = (a \cdot s) [(a \cdot b)s \wedge c - (a \cdot s)b \wedge c + (a \cdot c)b \wedge s] \\ & = (a \cdot s) (a \lrcorner (b \wedge s \wedge c)). \end{aligned}$$

Here the term containing $s \wedge s$ is the repeated-residual term that vanishes. Since $\text{sgn}(\{1\}) = 1$ and $\text{sgn}_S = 1$, this is exactly the sole-shared-seam identity of Section 2, including its sign convention.

There is an important qualification about vanishing. If a seam factor s_k is omitted from both selected sets, then the residual wedge contains $s_k \wedge s_k$ and is zero by alternation. However, a

term with $I \neq I_S$ need not vanish individually: its omitted seam factors may be complementary to those omitted in the other panel. Such nonprincipal terms are the T -summands in (3.2), and they combine by (3.3). It would therefore be incorrect to discard every $I \neq I_S$ term; only repeated-residual terms vanish before the minor convolution.

Finally, (3.1) applied to $b \wedge S \wedge c$, followed by (3.3), shows

$$\begin{aligned}(A \sqcup B) \wedge (A \sqcup C) &= (A \mid S)(A \sqcup (b \wedge S \wedge c)) \\ &= (A \mid S)(A \sqcup (B \sqcup C)).\end{aligned}$$

Restoring the orientation factors multiplies both sides by $\epsilon_B \epsilon_C$, which is already part of $B \sqcup C$. $\square$

Remark 2. *The grade count explains why the seam must have grade r . If $\text{grade}(A) = r$, $\text{grade}(S) = t$, and $\text{grade}(b) = p$, $\text{grade}(c) = q$, then the two sides before the coefficient have grades*

$$p + t - r + q + t - r \quad \text{and} \quad p + t + q - r,$$

respectively. Their difference is $t - r$. Thus a scalar coefficient is possible exactly when $t = r$. The vector proposition is the $r = t = 1$ case; the recursive coefficient uses the grade- r seam identity.

4 Nested Recursion

4.1 Orientation and parity

Before applying the seam identity, each panel is oriented so that the seam is rightmost in the left panel and leftmost in the right panel:

$$B = \epsilon_B b \wedge S, \quad C = \epsilon_C S \wedge c, \quad \epsilon_B, \epsilon_C \in \{+1, -1\}.$$

The factors ϵ_B and ϵ_C record the parity of these reorderings. They are part of the coefficient, so alignment is mathematical data rather than a cosmetic normalization step.

4.2 Sandhi and vichcheda

We borrow two terms from Sanskrit grammar. *Sandhi* (“joining”) is the stitch itself: replacing two panels that share a seam by one merged panel and a scalar, as in Theorem 1. *Vichcheda* (“splitting”) is the preparatory step that exposes the seam, and it acts on either side of a contraction.

Down vichcheda splits the two down arguments along their common visible factors S : it reorders them into the aligned forms $B = \epsilon_B b \wedge S$ and $C = \epsilon_C S \wedge c$ above, recording the parities.

Up vichcheda splits the contractors. If the two panels have contractors $K \wedge P$ and K , the extra factor P is peeled into the down argument,

$$(K \wedge P) \sqcup X = K \sqcup (P \sqcup X),$$

so both panels have the common contractor K , and $P \sqcup X$ becomes a candidate seam. When this seam has grade larger than $|K|$, the product vanishes (`upVichcheda_vanish`, Section 9).

The grade- r identity resolves the local seam step. The remaining nested construction is an iteration of vector contractions, not an application of the vector Leibniz rule to an accumulated blade. Write

$$\iota_u(X) := u \sqcup X.$$

For an ordered chain of vectors $d_1, \dots, d_r$, define

$$A_{\text{acc}}^{(r)} := d_1 \wedge d_2 \wedge \dots \wedge d_r,$$

$$\iota_{A_{\text{acc}}^{(r)}}(X) := \iota_{d_1}(\iota_{d_2}(\dots \iota_{d_r}(X) \dots)).$$

The recursive contraction rule

$$(u \wedge A) \lrcorner X = u \lrcorner (A \lrcorner X)$$

shows that this is exactly the corresponding blade contraction under the chosen ordering. Each local application of Theorem 1 is made with the current grade- r accumulated contractor and its grade- r accumulated seam. No blade-level Leibniz rule is being assumed.

At depth k , two nested panels may be written schematically as

$$T_L = \iota_{d_1}(X_1^L \wedge \iota_{d_2}(X_2^L \wedge \dots \iota_{d_k}(C_L) \dots)),$$

$$T_R = \iota_{d_1}(X_1^R \wedge \iota_{d_2}(X_2^R \wedge \dots \iota_{d_k}(C_R) \dots)).$$

When the recursive seam has accumulated to $S_{\text{acc}}^{(r)} = s_1 \wedge \dots \wedge s_r$, the coefficient

$$A_{\text{acc}}^{(r)} \mid S_{\text{acc}}^{(r)}$$

is scalar because both blades have grade r . The matching growth of the contractor and seam is therefore forced by grade, rather than an arbitrary convention.

At a given level, let

$$x = \text{grade}(D_L) + \text{grade}(D_R) - \text{grade}(D_{\text{merged}}) - \text{grade}(A_{\text{acc}}).$$

Under the sole-seam invariant, the three cases are:

$$\begin{array}{l|l} x < 0 & \text{recurse into the merged down expression,} \\ x = 0 & \text{apply the grade-}r \text{ identity and emit its scalar,} \\ x > 0 & \text{the expression vanishes because residual seam factors repeat.} \end{array}$$

The local grade- r seam identity is now a theorem. The remaining global claim concerns repeated application through arbitrary nested trees:

Inductive invariant. Let $h(\mathcal{T})$ be the number of unresolved recursive seam levels in a pair of nested panel trees. We use the following level-local invariant:

1. every down expression at a node is decomposable and homogeneous;
2. after orientation, the two panels have the form $B = b \wedge S$ and $C = S \wedge c$, with S their sole common visible factor;
3. a nested argument is opaque at its parent node and is inspected only when recursion reaches that argument's own node; and
4. all scalar factors and orientation parities produced above a node are carried unchanged into its merged child.

The word “visible” refers to immediate exterior-product arguments, not to the labelled leaves occurring inside nested contractions.

4.3 Level-local visibility

The closure argument uses a syntactic, rather than a global support, property. The panel grammar has three relevant constructors:

$$E ::= \text{atom} \mid E_1 \wedge \cdots \wedge E_n \mid (U \sqcup D).$$

At a fixed exterior-product node $D = E_1 \wedge \cdots \wedge E_n$, define

$$\text{Vis}(D) := \{E_1, \dots, E_n\}.$$

The seam matcher compares elements of $\text{Vis}(D_L)$ and $\text{Vis}(D_R)$; it does not compare arbitrary labelled leaves inside a nested contraction. A proper descendant of E_i is therefore opaque at this node. If the whole nested expression E_i occurs on both sides, it is one visible seam, and its own descendants are inspected only when recursion reaches that child node.

Lemma 3 (Syntactic visibility of seam candidates). *Let q be a proper descendant of an immediate factor E_i at an exterior node D . If $q \neq E_i$, then $q \notin \text{Vis}(D)$ and cannot be a seam candidate at D . A repeated nested expression E_i , by contrast, is one candidate at D ; its internal factors are candidates only at the nested node that contains them.*

Proof. By definition, $\text{Vis}(D)$ contains exactly the immediate arguments of the exterior node. A proper descendant of E_i is an argument of a child of E_i , not an argument of D , so it is not in $\text{Vis}(D)$. The seam matcher therefore cannot select it.

If the complete expression E_i occurs on both sides, the two immediate argument lists contain the same element E_i , so they expose one seam. The recursive matcher receives E_i as its own node only after descending to that node and then uses $\text{Vis}(E_i)$. No descendant is lifted into the parent argument list. $\square$

The following lemma supplies the algebraic bridge from visibility to the permitted expansion.

Lemma 4 (Semantic sibling separation). *Let $D = E_1 \wedge \cdots \wedge E_n$, where each E_i is either a one-blade or a nested contraction node, and let A be a blade of grade r . In the full multilinear expansion of $A \sqcup D$, expanding every nested node down to leaves, each monomial has the form*

$$\pm \Delta \cdot (F_1 \wedge \cdots \wedge F_n),$$

where F_i is a fully expanded residual built only from the leaves in the subtree E_i , and Δ is a product of pairings. No pairing term uses down-side leaves from two distinct sibling subtrees E_i and E_j .

Proof. For a flat exterior node, write $A = a_1 \wedge \cdots \wedge a_r$. Repeated application of the graded Leibniz rule partitions the ordered contractor factors among the siblings:

$$A \sqcup (E_1 \wedge \cdots \wedge E_n) = \sum_{I_1 \dot{\cup} \cdots \dot{\cup} I_n = [r]} \varepsilon(I_1, \dots, I_n) \bigwedge_{i=1}^n (A_{I_i} \sqcup E_i),$$

where A_{I_i} retains the original order. To make the sign explicit, let $\iota(j) = i$ when $j \in I_i$, let $g_i = \text{grade}(E_i)$, and put

$$\varepsilon(I_1, \dots, I_n) := (-1)^{\sum_{j=1}^r \sum_{k < \iota(j)} g_k + \#\{(j, \ell) : j < \ell, \iota(j) > \iota(\ell)\}}.$$

The first term is the sum of the grades crossed by the j -th contraction; the second is the inversion parity created when an earlier contraction lowers the grade crossed by a later one. For a one-blade E_i , the factor is zero unless the assigned grade is admissible, and its nonzero coefficient is the corresponding dot pairing. For a nested E_i , the same expression is expanded recursively

inside that factor. Induction on the total number of nested contraction nodes therefore expands each $A_{I_i} \sqcup E_i$ independently. All pairings created there have both their down-side leaves inside E_i ; wedging the residuals introduces only signs and possible repeated-factor zeros, never a new pairing across siblings. This proves the claim. $\square$

The syntactic visibility lemma is therefore a statement about what the matcher can select, while semantic sibling separation supplies the algebraic bridge. Together they justify ignoring a buried coincidence without defining it away: Theorem 2 proves that the permitted stitches still equal full direct left-contraction expansion. In particular, this covers the hidden-coincidence configuration

$$D_L = a_2 \wedge a_3 \wedge (a_4 \sqcup (b_1 \wedge b_2)), \quad D_R = a_3 \wedge b_1,$$

which is also checked by the corresponding implementation regression test.

Lemma 5 (One-stitch closure). *Let T_L, T_R be well-formed nested contraction trees of height h satisfying the level-local sole-seam invariant and sharing a contraction lineage. For the stitch defined in the Algorithm section, grammar preservation follows directly from its definition: it concatenates immediate exterior arguments and recursively visits a child only at that child's own node. If T_M is the result of one valid root stitch, then T_M satisfies the same invariant at every remaining level.*

Proof. Induct on h . For $h = 0$, there are no remaining levels.

For $h > 0$, orient the root down expressions as

$$D_L = b \wedge S, \quad D_R = S \wedge c.$$

The stitch constructs

$$D_M = b \wedge S \wedge c,$$

with S counted once. At this node, the only possible seam candidates come from the immediate argument lists. Lemma 3 excludes every factor buried inside a nested argument. Thus a factor of c that coincides with a buried factor of T_L cannot create a second seam at the root. If it coincides with a complete immediate nested argument, that complete argument is visible and the original sole-seam hypothesis already excludes a second candidate.

By Lemma 4 and the definition of the stitch, c is not paired with a buried leaf of b 's nested child, nor is a child factor lifted into the parent list. Hence every remaining comparison is either the unique corresponding comparison inside S , or a comparison at a separate child node whose immediate arguments were already governed by the sole-seam invariant. The former has strictly smaller height and is preserved by the induction hypothesis; the latter cannot acquire a new cross-level candidate by Lemma 3.

Therefore the merged tree has exactly one active seam at every remaining lineage level. Re-ordering signs and scalar overlap coefficients change coefficients only, not the syntax or visibility relation. $\square$

The earlier hereditary-freshness condition is sufficient but stronger than necessary: labelled leaves may coincide across levels, as long as nested constructions remain opaque to seam matching at their parent node. The lemma therefore applies unconditionally to the stated well-formed panel grammar.

Theorem 2 (Nested sole-seam recursion under the invariant). *For a fixed valid merge schedule, any pair of well-formed nested contraction trees satisfying the level-local sole-seam invariant is reduced by repeated grade-gated stitching to the same expression as direct left-contraction expansion. The result includes all overlap scalars and exterior-algebra parity signs.*

Proof. We induct on $h(\mathcal{T})$. If $h(\mathcal{T}) = 0$, the current seam is grade-matched with the accumulated contractor, so $x = 0$. The grade- r sole-seam identity, Theorem 1, gives exactly the direct contraction expansion, including the orientation factor in $B \sqcup C$.

Now assume the claim for all trees of height at most h , and consider a tree of height $h + 1$. Orient the root panels and expand each contraction by the graded Leibniz rule. If $x > 0$, Lemma 4 says that contractor factors assigned to one sibling cannot remove a seam factor through a pairing with a buried leaf in another sibling. Since the visible seam has grade larger than the available contractor grade, every summand retains a common seam factor in both residual wedges. Each such summand is zero by alternation, and the grade-gated recursion returns zero as well.

If $x = 0$, the current contractor and seam have equal grade. The root is therefore exactly the grade- r situation of Theorem 1; replacing the two panels by their merged panel preserves the direct expansion and extracts the scalar $A_{\text{acc}} \mid S_{\text{acc}}$.

It remains to consider $x < 0$. The graded Leibniz expansion groups the nonzero terms by the unique residual visible seam. By the one-stitch closure lemma, this group is represented by a merged down-expression of height at most h , with exactly the scalar and parity factors recorded at the root. The induction hypothesis identifies the recursive reduction of that merged expression with its direct left-contraction expansion. Multiplying by the unchanged root factors gives the direct expansion of the original tree. This proves the claim for height $h + 1$, and hence for every finite height. $\square$

The theorem is deliberately stated for a fixed valid merge schedule. The closure lemma establishes its level-local structural hypothesis for the well-formed panel grammar. The theorem does not claim that different valid schedules are confluent. Computational checks of nesting depths up to six, including higher-grade decomposable blades, support the invariant but do not replace its proof.

4.4 Closed form for a nest with an innermost seam

When the shared seam sits at the bottom of a contraction chain, the recursion of Theorem 2 can be written in one formula. Let $K_1, \dots, K_k$ be contractor blades and U_i, V_i wing blades, and let U_x, S, W be blades. Put

$$\begin{aligned}\Phi^L &:= K_1 \lrcorner (U_1 \wedge K_2 \lrcorner (U_2 \wedge \dots \wedge K_k \lrcorner (U_k \wedge U_x \wedge S))), \\ \Phi^R &:= K_1 \lrcorner (V_1 \wedge K_2 \lrcorner (V_2 \wedge \dots \wedge K_k \lrcorner (V_k \wedge S \wedge W))), \\ \Phi^M &:= K_1 \lrcorner (U_1 \wedge V_1 \wedge \dots \wedge K_k \lrcorner (U_k \wedge V_k \wedge U_x \wedge S \wedge W)),\end{aligned}$$

and $A_{\text{acc}} := K_1 \wedge \dots \wedge K_k$.

Theorem 3 (Nested sole-seam identity, closed form). *If $\text{grade}(A_{\text{acc}}) = \text{grade}(S)$, then*

$$\Phi^L \wedge \Phi^R = (-1)^\sigma (A_{\text{acc}} \mid S) \Phi^M, \quad \sigma = \sum_{i=1}^k |V_i| \left(|U_x| + \sum_{j>i} |U_j| \right),$$

where $|\cdot|$ is grade. If $\text{grade}(A_{\text{acc}}) < \text{grade}(S)$, then $\Phi^L \wedge \Phi^R = 0$. No disjointness or nondegeneracy hypothesis is needed.

Proof sketch. Prove the stronger statement in which Φ^R is replaced by $E \lrcorner \Phi^R$ for an extra contractor E , by induction on k . At each level the outer K_1 annihilates Φ^L and is pulled out of both factors; the accumulated contractor $E \wedge K_1$ is then pushed past V_1 , and every error term is a deficient instance, hence zero. At the bottom, $S \wedge (E \lrcorner (S \wedge Y)) = S \wedge (E \lrcorner S) \wedge Y$, and $E \lrcorner S = (E \mid S)$ by the Capelli extraction. The sign σ is the shuffle parity of moving each V_i past the deeper left wings. The full proof is machine-checked (Section 9). $\square$

For $k = 1$ this is Theorem 1. The identity

$$\begin{aligned} & (d_1 \lrcorner (a_1 a_2 \wedge d_2 \lrcorner (b_1 b_2 \wedge d_3 \lrcorner S))) \wedge (d_1 \lrcorner (a_3 a_4 \wedge d_2 \lrcorner (b_3 b_4 \wedge d_3 \lrcorner (S \wedge c_4)))) \\ &= (d_1 d_2 d_3 \mid S) d_1 \lrcorner (a_1 a_2 a_3 a_4 \wedge d_2 \lrcorner (b_1 b_2 b_3 b_4 \wedge d_3 \lrcorner (S \wedge c_4))), \quad S = c_1 c_2 c_3, \end{aligned}$$

with juxtaposition denoting $\wedge$, is the case $k = 3$, $\sigma = 4$.

4.5 The overlap coefficient via Capelli

At a balanced grade gate, the coefficient is the scalar pairing of the accumulated contractor and accumulated seam:

$$A_{\text{acc}} \mid S_{\text{acc}} = (-1)^{\binom{r}{2}} \det[(d_i \cdot s_j)]_{i,j=1}^r.$$

The factor $(-1)^{\binom{r}{2}}$ is the reversal forced by the iterated contraction order of Section 1: for $\text{grade}(A) = \text{grade}(S)$, $A \lrcorner S$ is exactly this scalar. This determinant is the finite Capelli contraction of the accumulated contractor with the grade-matched seam. Reordering parities are multiplied into the coefficient separately.

5 Algorithm (Stitching)

The rewrite is described abstractly:

```
canonicalize(expression):
  while a mergeable pair exists:
    choose two panels with a common contraction lineage
    orient the panels around one shared seam
    record the two reordering parities
    split their down blades into unique parts and one seam
    accumulate the contractor and seam blades
    compute the grade-balance value x
    if x > 0:
      replace the expression by zero under the sole-seam invariant
    else if x < 0:
      recursively canonicalize the merged down expression
    else:
      multiply by the scalar contractor-seam pairing and both parities
      replace the pair by the merged contraction
  return expression
```

6 CGA encoding

The applications use the standard conformal model rather than introducing a new encoding. For $x \in \mathbb{R}^n$, its conformal point is the null vector

$$\dot{x} = x + e_0 + \frac{x^2}{2} e_\infty, \quad \dot{x}^2 = 0,$$

with the usual normalization $e_0 \cdot e_\infty = -1$. A circle or sphere is represented by the blade obtained from its defining points or carrier objects. We write $\vee$ for the standard dual/reduced meet. Incidence is expressed by

$$A \wedge \dot{y} = 0.$$

For a sphere with Euclidean centre c and radius ρ , the dual representation is

$$\dot{c} - \frac{\rho^2}{2} e_\infty.$$

These conventions, including lines, circles, spheres, duality, and incidence, are standard; see Li [1], Doran and Lasenby [5], and Dorst, Fontijne, and Mann [6]. The present paper only applies stitching to the resulting exterior-product and contraction expressions.

6.1 Radius-augmented Lie-sphere chart

The paper uses two representation models. The preceding point-only conformal model is used for point, line, circle, and sphere incidence. For oriented circles in the Monge application, we additionally use the radius-augmented Lie-sphere chart $\mathbb{R}^{3,2}$, following the standard Lie-sphere framework [2]. Take basis $e_0, e_x, e_y, e_r, e_\infty$ with

$$e_0 \cdot e_\infty = -1, \quad e_x^2 = e_y^2 = 1, \quad e_r^2 = -1,$$

and all other products zero. An oriented circle with centre (x, y) and signed radius r is

$$\mathcal{C}(x, y, r) := e_0 + xe_x + ye_y + re_r + \frac{x^2 + y^2 - r^2}{2} e_\infty.$$

It is null, since

$$\mathcal{C} \cdot \mathcal{C} = x^2 + y^2 - r^2 - (x^2 + y^2 - r^2) = 0.$$

The e_r -coordinate carries the signed radius. This chart is used only where the radius coordinate is geometrically relevant; the two models share the same exterior and contraction operations.

7 Applications

The step counts below count successful seam-resolution steps, not elementary geometric constructions, scalar products, or final incidence tests.

7.1 A parameterized two-stitch incidence family

The Monge, perpendicular-bisector, and angle-bisector constructions have the same algebraic shape when their chart encodings satisfy the same panel hypotheses. Let X_1, X_2, X_3 be oriented carrier blades and let K be a fixed contractor. Put

$$Q_{ij} := K \lrcorner (X_i \wedge X_j).$$

Assume that the three panels are oriented so that Q_{12} and Q_{23} have the sole seam X_2 , that $Q_{12} \wedge Q_{23}$ exposes the two-factor seam $X_3 \wedge X_1$ against Q_{31} , and that $\text{grade}(K) = \text{grade}(X_2) = 1$. The same two-stitch proof then applies to every geometric interpretation of the carriers.

Proposition 2 (Parameterized pairwise-difference incidence). *Under the preceding panel hypotheses, the three pairwise objects Q_{12} , Q_{23} , and Q_{31} satisfy*

$$Q_{12} \wedge Q_{23} \wedge Q_{31} = 0.$$

Consequently, whenever the resulting blades represent the relevant affine objects in a common chart, their three pairwise constructions have the corresponding common incidence or collinearity relation.

Proof. The first seam is grade-matched, so the grade- r identity gives

$$Q_{12} \wedge Q_{23} = (K \mid X_2) K \lrcorner (X_1 \wedge X_2 \wedge X_3),$$

up to the canonical orientation sign already fixed by the panel order. Against Q_{31} , the exposed seam is $X_3 \wedge X_1$, of grade two, whereas K has grade one. The grade gate is therefore in

its overflow branch: every full-expansion term retains a repeated seam factor and vanishes by alternation. Hence

$$Q_{12} \wedge Q_{23} \wedge Q_{31} = 0.$$

The proof depends only on the contractor grade, seam pattern, and orientation; the geometric meaning of X_i is the parameter. $\square$

The intended instantiations are summarized below. The first row is realized in the Lie-sphere chart of this paper. The other two rows are algebraic targets for a chart-specific encoding, not claims that the point-only CGA chart automatically supplies those panels.

construction	K	carriers	pairwise result	status here
Monge	e_r	oriented circles	homothety centre	proved
perpendicular bisectors	e_∞	points	perpendicular bisector	encoding required
angle bisectors	e_r	lines	angle bisector	encoding required

Thus the proposition expresses the common two-stitch architecture, while the carrier chart must still be checked for each construction. In particular, the often-suggested line representative

$$L = xe_x + ye_y + e_r + de_\infty, \quad x^2 + y^2 = 1,$$

is null in the radius-augmented metric, but it does not make the angle-bisector row valid by itself. For

$$L_1 = e_x + e_r + 2e_\infty, \quad L_2 = e_y + e_r - e_\infty,$$

the direct metric calculation gives

$$L_1^2 = L_2^2 = 0, \quad e_r \lrcorner (L_1 \wedge L_2) = L_1 - L_2, \quad (L_1 - L_2)^2 = 2 \neq 0.$$

The contracted blade is therefore not null and cannot be identified with a null line in this naive chart. This numerical check prevents the angle- and perpendicular-bisector rows from being overstated; a different representation, or a separate metric lemma, is required before they become fully verified CGA applications.

Verified instantiation: Monge. We now use the Lie-sphere chart introduced above. The sign of r records orientation; reversing it exchanges the internal and external homothety interpretations.

Lemma 6 (Radius annihilator). *For two oriented circles $\mathcal{C}_i, \mathcal{C}_j$, put*

$$P_{ij} := e_r \lrcorner (\mathcal{C}_i \wedge \mathcal{C}_j).$$

Then

$$P_{ij} = (e_r \cdot \mathcal{C}_i)\mathcal{C}_j - (e_r \cdot \mathcal{C}_j)\mathcal{C}_i = r_j\mathcal{C}_i - r_i\mathcal{C}_j.$$

Its e_r -coefficient is zero, and its e_0 -coefficient is $r_j - r_i$. If $r_i \neq r_j$, its normalized Euclidean centre is

$$\frac{r_j\mathbf{c}_i - r_i\mathbf{c}_j}{r_j - r_i},$$

the homothety centre of the two circles. The vector P_{ij} need not itself be null; the relevant homogeneous point coordinates are its (e_0, e_x, e_y) -components.

Proof. The first equality is the vector Leibniz rule, and $e_r \cdot \mathcal{C}_i = -r_i$. The e_r -coefficient of $r_j\mathcal{C}_i - r_i\mathcal{C}_j$ is $r_jr_i - r_i r_j = 0$. Dividing its e_0 -normalized spatial part by $r_j - r_i$ gives the displayed centre. $\square$

Theorem 4 (Monge). *For three oriented circles in general position, the three homothety centres P_{12}, P_{23}, P_{31} are collinear.*

Proof. First stitch the panels P_{12} and P_{23} . Their down blades have the sole visible seam $\mathcal{C}_2$, already oriented as

$$\mathcal{C}_1 \wedge \mathcal{C}_2, \quad \mathcal{C}_2 \wedge \mathcal{C}_3.$$

The contractor e_r and seam $\mathcal{C}_2$ both have grade one, so the balanced grade- r identity gives

$$\begin{aligned} P_{12} \wedge P_{23} &= (e_r \cdot \mathcal{C}_2) e_r \lrcorner (\mathcal{C}_1 \wedge \mathcal{C}_2 \wedge \mathcal{C}_3) \\ &= -r_2 e_r \lrcorner (\mathcal{C}_1 \wedge \mathcal{C}_2 \wedge \mathcal{C}_3). \end{aligned}$$

The resulting panel and P_{31} now expose the two-factor seam

$$S = \mathcal{C}_3 \wedge \mathcal{C}_1,$$

while the accumulated contractor remains e_r . Hence $\text{grade}(S) - \text{grade}(e_r) = 2 - 1 > 0$. The grade gate is therefore in its vanishing branch: every term in the full expansion retains a repeated residual seam factor, and alternation makes it zero. Consequently,

$$P_{12} \wedge P_{23} \wedge P_{31} = 0.$$

Projecting this linear dependence to the (e_0, e_x, e_y) -coordinates gives the corresponding dependence of the three homogeneous Euclidean point coordinates. Since the e_0 -coefficients are nonzero in general position, their normalized centres are collinear. $\square$

The signed-radius representation covers the orientation cases uniformly. Circles with radii of the same sign yield the external homothety centre, whereas opposite signs yield the internal one. The proof does not inspect these signs: it consists of one balanced stitch followed by one grade overflow. Thus the same two-step reduction proves the classical external case and all mixed internal/external cases realizable by a consistent signed orientation. The all-internal choice is not generally collinear and is not reachable by any consistent choice of signed radii. For example, with centres $(0, 0), (4, 0), (1, 3)$ and signed radii $(-1, -2, -3)$, the three centres are $(-4, 0), (10, -6), (-\frac{1}{2}, -\frac{3}{2})$, whose affine collinearity determinant is zero. In contrast, forcing the three pairwise internal centres for the corresponding positive radii gives determinant $12/5$, confirming that the all-internal choice is not a theorem covered by signed orientation.

The reduction uses two successful seam-resolution steps: one balanced merge emitting the scalar $e_r \cdot \mathcal{C}_2 = -r_2$, followed by one grade overflow. The largest intermediate is the grade-three circle wedge $\mathcal{C}_1 \wedge \mathcal{C}_2 \wedge \mathcal{C}_3$. The assumptions exclude equal-radius centres at infinity and the usual linearly dependent degeneracies.

7.2 Concyclic four-point boundary case

The concyclic four-point case uses a different termination mechanism and is not an instance of Proposition 2. For null point representatives $\dot{p}_1, \dots, \dot{p}_4$, concyclicity is the incidence condition

$$\dot{p}_1 \wedge \dot{p}_2 \wedge \dot{p}_3 \wedge \dot{p}_4 = 0.$$

Here the computation terminates because the four point representatives are linearly dependent, not because an accumulated seam has exceeded the contractor grade. This distinction matters: the example shows that the stitching calculus uses geometric hypotheses such as concyclicity in addition to grade balance. We therefore record it as a boundary case rather than claim a two-stitch contraction proof for it.

8 Related work

Li’s 2026 treatment of null geometric algebra studies null monomials, centred null binomials, and their geometric interpretation. In particular, equation (5.1) gives a factorization rule for products of anisotropic null monomials with a common null endpoint, reducing the product to versor factors and endpoint factors [1]. That is a null-monomial/geometric-product normal-form statement. The identity proved here is different: it is the local exterior-contraction rewrite $(A \lrcorner B) \wedge (A \lrcorner C)$, with a grade-dependent overlap scalar and explicit seam parity.

Li, Shao, Huang, and Liu’s 2014 work on reduction among bracket polynomials develops an invariant division and an admissible order leading to an invariant Gröbner-basis theory [3]. We checked the available statement and proceedings description for an overlap with the present construction. Its invariant division reduces bracket polynomials modulo an algebraic ordering; it does not state the local contraction-panel rewrite or the sole-shared-seam identity proved here. The two approaches are therefore complementary: invariant division addresses global polynomial reduction, whereas *sandhi* addresses a local geometric contraction pattern.

The 2008 monograph *Invariant Algebras and Geometric Reasoning* contains the relevant null-bracket-algebra and Euclidean-geometry chapters, alongside Grassmann–Cayley, conformal, and Clifford-algebra material [4]. Our check was limited to the available table of contents, bibliographic records, searchable previews, and descriptions of the “Null Bracket Algebra and Euclidean Geometry” material; full access to the book was not available. Those sources establish the surrounding bracket algebra context, but they were insufficient to verify every chapter-level identity, so no claim of exhaustive literature exclusion is made.

9 Formal verification and implementation

The stitching identities are proved in Lean 4 (v4.33.1) with Mathlib, in `paper/sandhi/lean/` of the Sygal repository, <https://github.com/Akash-dvd/sygal/tree/main/paper/sandhi/lean>, with no unproved assumptions beyond Lean’s standard axioms. The setting is ΛM for a module M over an arbitrary commutative ring, with left contraction by an arbitrary bilinear form; in particular, neither nondegeneracy nor disjointness of wing factors is required.

Machine-checked result	Lean name
Base identity (Section 2)	<code>soleSeam_grade1</code>
Grade- r identity, Theorem 1	<code>soleSeam_gradeR</code>
Flat overflow (grade $S > \text{grade } A$) gives 0	<code>soleSeam_vanish</code>
Capelli: $A \lrcorner S = (-1)^{\binom{r}{2}} \det[a_i \cdot s_j]$	<code>contractBlade_ofList_eq_seamPairing</code>
Nested closed form, Theorem 3	<code>nested_soleSeam'</code>
Nested overflow gives 0	<code>nested_vanish</code>
Up vichcheda: $(K \lrcorner S) \wedge K \lrcorner (w_1 \wedge S \wedge w_2) = 0$	<code>upVichcheda_vanish</code>
Expansion (3.1) for $r = 1$	<code>contractVec_ofList_eq_expandSum</code>

In the up-vichcheda row the contractors are $K \wedge P$ and K , the seam is the peeled panel $S = P \lrcorner X$, and grade $S > |K|$; the result holds over a field, for arbitrary wings, because $P \lrcorner X$ is again a blade when X is a wedge of vectors (`contractBlade_ofList_blade`). It is false for a general homogeneous S .

The following have paper proofs only: the expansion (3.1) for $r > 1$ and the sign lemma for the minor convolution; the canonical-orientation lemma; the visibility, sibling-separation, and one-stitch closure lemmas of Section 4; the nested recursion theorem (Theorem 2) for seams distributed across levels; and the CGA encoding and the results of the Applications section (the incidence proposition, the radius annihilator, and Monge’s theorem). Schedule independence is open.

The Lean proof of the grade- r identity does not use the minor expansion of Section 3. It uses self-annihilation (each contractor factor annihilates $A \lrcorner (b \wedge S)$), the fact that any term spending a contractor on b leaves a seam factor that dies against S , and a Laplace expansion along the last row for the Capelli coefficient.

The implementation (the Sygal canonicalizer) does two things: it searches an expression for panels sharing a seam and orients them, and it then replaces the pair by the right-hand side of the applicable theorem. Only the second step is a mathematical claim, and it is the one covered above. The search is pattern matching; its correctness amounts to applying the theorem with the correct seam, contractors, orientation parities, and coefficient.

An exact cross-check against the Lean formulas (`tests/crosscheck_lean.py`, evaluating both sides with rational vectors and random symmetric forms) finds that the implementation's pairing and Capelli coefficient equal $(-1)^{\binom{r}{2}} \det[k_i \cdot s_j]$ for $r \leq 4$, and that each flat stitch it performs equals the right-hand side of Theorem 1, sign included; its up-vichcheda reductions return 0 exactly in the cases covered by `upVichcheda_vanish`. The implementation does not yet apply Theorem 3: when the outer levels share no visible factor it leaves the product unchanged, which is correct but not reduced. The three-level identity after Theorem 3 is therefore checked by expansion and by Lean, not by the canonicalizer.

10 Limitations

The grade- r seam identity is machine-checked for arbitrary blades given as wedge products of vectors; the sole-seam hypotheses are needed only to interpret the result as a unique stitch. The fixed-schedule induction theorem is proved for the well-formed, level-local panel grammar; schedule-independent normalization remains open. Since panel orientation and its parity are canonical once the atom order is fixed, the remaining source of schedule dependence is pair selection: which eligible seam is stitched first. Cases through nesting depth six, including higher-grade decomposable blades, have been verified computationally. The nested identity is machine-checked for a seam at the innermost level of a shared contractor chain (Theorem 3); configurations in which seam factors are distributed across levels rely on the paper proof of Theorem 2. The paper does not claim a normal-form theorem for arbitrary mixed-grade or nondecomposable multivectors.

References

- [1] H. Li, “Null geometric algebra with geometric interpretation,” *Philosophical Transactions of the Royal Society A*, 384:20250096, 2026. <https://doi.org/10.1098/rsta.2025.0096>.
- [2] T. E. Cecil, *Lie Sphere Geometry: With Applications to Submanifolds*, Springer-Verlag, 1992.
- [3] H. Li, C. Shao, L. Huang, and Y. Liu, “Reduction among bracket polynomials,” in *Proceedings of ISSAC 2014*, pp. 304–311, 2014. <https://doi.org/10.1145/2608628.2608630>.
- [4] H. Li, *Invariant Algebras and Geometric Reasoning*, World Scientific, 2008. <https://doi.org/10.1142/6514>.
- [5] C. Doran and A. Lasenby, *Geometric Algebra for Physicists*, Cambridge University Press, 2003.
- [6] L. Dorst, D. Fontijne, and S. Mann, *Geometric Algebra for Computer Science*, Morgan Kaufmann, 2007.

- [7] F. R. Gantmacher, *The Theory of Matrices*, Vol. 1, Chelsea Publishing Company, 1959, Chapter I, §5.